

Assessment (non-exam) Brief

Module code/name	MSIN0143
Module leader name	David Alderton
Academic year	2023/24
Term	1
Assessment title	Building an Analytical Notebook in Python
Individual/group assessment	Group

Submission deadlines: Students should submit all work by the published deadline date and time. Students experiencing sudden or unexpected events beyond your control which impact your ability to complete assessed work by the set deadlines may request mitigation via the [extenuating circumstances procedure](#). Students with disabilities or ongoing, long-term conditions should explore a [Summary of Reasonable Adjustments](#).

Return and status of marked assessments: Students should expect to receive feedback within one calendar month of the submission deadline, as per UCL guidelines. The module team will update you if there are delays through unforeseen circumstances (e.g. ill health). All results when first published are provisional until confirmed by the Examination Board.

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Academic Misconduct: Academic Misconduct is defined as any action or attempted action that may result in a student obtaining an unfair academic advantage. **Academic misconduct includes plagiarism, obtaining help from/sharing work with others be they individuals and/or organisations or any other form of cheating.** Refer to [Academic Manual Chapter 6, Section 9: Student Academic Misconduct Procedure - 9.2 Definitions](#).

Referencing: You must reference and provide full citation for ALL sources used, including AI sources, articles, text books, lecture slides and module materials. This includes any direct quotes and paraphrased text. If in doubt, reference it. If you need further guidance on referencing please see [UCL's referencing tutorial for students](#). Failure to cite references correctly may result in your work being referred to the Academic Misconduct Panel.

Use of Artificial Intelligence (AI) Tools in your Assessment: Your module leader will explain to you if and how AI tools can be used to support your assessment. In some assessments, the use of generative AI is **not permitted** at all. In others, AI may be used in an **assistive** role which means students are permitted to use AI tools to support the development of specific skills required for the assessment as specified by the module leader. In others, the use of AI tools may be an **integral** component of the assessment; in these cases the assessment will provide an opportunity to demonstrate effective and responsible use of AI. See page 3 of this brief to check which category use of AI falls into for this assessment. Students should refer to the [UCL guidance on acknowledging use of AI and referencing AI](#). Failure to correctly reference use of AI in assessments may result in students being reported via the Academic Misconduct procedure. Refer to the section of the UCL Assessment success guide on [Engaging with AI in your education and assessment](#).

Content of this assessment brief

Section	Content
A	Core information
B	Coursework brief and requirements
C	Module learning outcomes covered in this assessment
D	Groupwork instructions (if applicable)
E	How your work is assessed
F	Additional information

Section A: Core information

Submission date	12/12/2023
Submission time	10.00am
Assessment is marked out of:	100
% weighting of this assessment within total module mark	40%
Maximum word count/page length/duration	2000 words
Footnotes, appendices, tables, figures, diagrams, charts included in/excluded from word count/page length?	excluded in word count
Bibliographies, reference lists included in/excluded from word count/page length?	excluded in word count
Penalty for exceeding word count/page length	Penalty for exceeding word count will be a deduction of 10 percentage points, capped at 40% for Levels 4,5, 6, and 50% for Level 7) Refer to Academic Manual Section 3: Module Assessment - 3.13 Word Counts.
Penalty for late submission	Standard UCL penalties apply. Students should refer to https://www.ucl.ac.uk/academic-manual/chapters/chapter-4-assessment-framework-taught-programmes/section-3-module-assessment#3.12
Artificial Intelligence (AI) category	Assistive
Submitting your assessment	The assignment MUST be submitted to the module submission link located within this module's Moodle 'Submissions' tab by the specified deadline. Where the assessment is group work, each group should nominate ONE (1) person to upload the submission on behalf of their group. Do NOT submit more than one submission per group.
Anonymity of identity. Normally, <u>all</u> submissions are anonymous unless the nature of the submission is such that anonymity is not appropriate, illustratively as in presentations or where minutes of group meetings are required as part of a group work submission	The nature of this assessment is such that anonymity is not required.

Section B: Assessment Brief and Requirements

This group coursework is intended to allow you to demonstrate the programming skills that you have developed during the course.

The group should produce a Jupyter Notebook that performs an **exploratory analysis** of a dataset that you have chosen. The Notebook should contain Markup text to explain your analysis and comprise 2000 words. In practise this might be 20 cells of Markup containing approximately 100 words.

You should convert the Notebook to pdf for the purposes of submitting to Moodle. You will also upload the Notebook to a shared platform (TBD) that allows the markers to review the code and the analysis in a live environment.

You should follow a **similar** structure of the analysis presented in Lecture 5, based on Chapter 2 of *Hands-on Machine Learning with Scikit-Learn and Tensorflow*, Aurélien Géron.

You need to go as far as building models and testing and evaluating their performance. You will be able to get full credit by demonstrating a thorough exploratory analysis of your chosen dataset.

To help guide your thinking you can use the following adapted version of the Machine Learning Project Checklist FROM Geron, A. *Hands-On Machine Learning with Scikit-Learn and Tensor Flow*: O'Reilly Media Inc 2017

A. Get the Data

1. Find and document where you can get data.
2. Get access authorizations.
3. Create a workspace (with enough storage space).
4. Retrieve the data.
5. Convert the data to a format you can easily manipulate (without changing the data itself).
6. Ensure sensitive information is deleted or protected (e.g., anonymized).
7. Check the size and type of data.

B. Explore the Data

1. Create a copy of the data for exploration (sampling it down to a manageable size if necessary).
2. Create a master Jupyter notebook to keep a record of your data exploration.
3. Study each attribute and its characteristics:
 - Name
 - Type (categorical, int/float, bounded/unbounded, text, structured, etc.)
 - % of missing values
 - Noisiness and type of noise (stochastic, outliers, rounding errors, etc.)
 - Possibly useful for the task?
 - Type of distribution (Gaussian, uniform, logarithmic, etc.)
4. For supervised learning tasks, identify the target attribute(s).
5. Visualize the data.
6. Study the correlations between attributes.
7. Study how you would solve the problem manually.
8. Identify the promising transformations you may want to apply.

9. Identify extra data that would be useful.
10. Document what you have learned.

C. Prepare the Data

1. Work on copies of the data, keeping the original dataset intact.
2. Write functions for all data transformations you can apply next time you get a fresh dataset.
3. Fix or remove outliers.
4. Fill in missing values or drop their rows or columns.
5. Drop the attributes that provide no useful information for the task.
6. Where appropriate:
 - Discretize continuous features.
 - Decompose features (e.g., categorical, date/time, etc.).
 - Add promising transformations of features (e.g., $\log(x)$, $\sqrt{x}$, x^2 , etc.).
 - Aggregate features into promising new features.
7. Standardize or normalize features.

D. Short-List Promising Models

1. Train at least 2 models using standard parameters.
2. Measure and compare their performance.
3. Analyze the most significant variables for each algorithm.
4. Analyze the types of errors the models make.
5. Short-list the most promising model

E. Present Your Analysis

1. Document what you have done.
2. Create a readable Jupyter Notebook.
3. Make sure you highlight the big picture first.
4. Don't forget to present interesting points you noticed along the way.
5. Describe what worked and what did not.
6. List your assumptions and your system's limitations.
7. Ensure your key findings are communicated through impactful visualisations or easy-to-remember statements.

Section C: Module Learning Outcomes covered in this Assessment

This assessment contributes towards the achievement of the following stated module Learning Outcomes as highlighted below:

1. Be able to explore data and program simple algorithms
2. Evaluate computational approaches to data analytics problems
3. Visualise data for business analytics

Section D: Groupwork Instructions (where relevant/appropriate)

This is a group work assessment and the following associated instructions apply:

1. Groups for this coursework are in teams of three (3) four (4) or five (5)
2. ALL members of the Team are responsible for ALL parts of the submission. It is your Team's responsibility to submit a coherent piece of work that ALL members can legitimately verify as a group effort.
3. You should manage your project through a TRELLO board.
4. All meetings should be recorded in TRELLO and specify at a minimum
 - a. attendance, and absentees
 - b. agreements as to who was allocated which tasks
 - c. reviews of work done by individual members
5. The Team mark applies to all Team members who contributed equitably to the final submission.
6. If there are issues within the group you MUST inform your module leader as early as possible during the coursework preparation, to enable any required interventions to be made. Note these issues within TRELLO.
7. Once the coursework has been submitted, if no prior notification of group/team challenges involving non-contributors has been made, complaints after submission will not normally be entertained.

Section E: How your work is assessed

Within each section of this assessment you may be assessed on the following aspects, as applicable and appropriate to this assessment, and should thus consider these aspects when fulfilling the requirements of each section:

- The accuracy of any calculations required.
- The strengths and quality of your overall analysis and evaluation;
- Appropriate use of relevant theoretical models, concepts and frameworks;
- The rationale and evidence that you provide in support of your arguments;
- The credibility and viability of the evidenced conclusions/recommendations/plans of action you put forward;
- Structure and coherence of your considerations and reports;
- Appropriate and relevant use of, as and where relevant and appropriate, real world examples, academic materials and referenced sources. Any references should use either the Harvard OR Vancouver referencing system (see [References, Citations and Avoiding Plagiarism](#))
- Academic judgement regarding the blend of scope, thrust and communication of ideas, contentions, evidence, knowledge, arguments, conclusions.
- Each assessment requirement(s) has allocated marks/weightings.

Student submissions are reviewed/scrutinised by an internal assessor and are available to an External Examiner for further review/scrutiny before consideration by the relevant Examination Board.

It is not uncommon for some students to feel that their submissions deserve higher marks (irrespective of whether they actually deserve higher marks). To help you assess the relative strengths and weaknesses of your submission please refer to [SOM Assessment Criteria Guidelines](#), located on the Assessment tab of the SOM Student Information Centre Moodle site.

The above is an important link as it specifies the criteria for attaining the pass/fail bandings shown below:

At UG Levels 4, 5 and 6:

80% to 100%: Outstanding Pass - 1st; 70% to 79%: Excellent Pass - 1st; 60%-69%: Very Good Pass - 2.1; 50% to 59%: Good Pass - 2.2; 40% to 49%: Satisfactory Pass - 3rd; 20% to 39%: Insufficient to Pass - Fail; 0% to 19%: Poor and Insufficient to Pass - Fail.

At PG Level 7:

86% to 100%: Outstanding Pass - Distinction; 70% to 85%: Excellent Pass - Distinction; 60%-69%: Good Pass - Merit; 50% to 59%: Satisfactory - Pass; 40% to 49%: Insufficient to Pass - Fail; 0% to 39%: Poor and Insufficient to Pass - Fail.

You are strongly advised to review these criteria before you start your work and during your work, and before you submit.

You are strongly advised to **not** compare your mark with marks of other submissions from your student colleagues. Each submission has its own range of characteristics which differ from others in terms of breadth, scope, depth, insights, and subtleties and nuances. On the surface one submission may appear to be similar to another but invariably, digging beneath the surface reveals a range of differing characteristics.

Students who wish to request a review of a decision made by the Board of Examiners should refer to the [UCL Academic Appeals Procedure](#), taking note of the [acceptable grounds](#) for such appeals.

Note that the purpose of this procedure is not to dispute academic judgement – it is to ensure correct application of UCL's regulations and procedures. The appeals process is evidence-based and circumstances must be supported by independent evidence.

Section F: Additional information from module leader (as appropriate)

MSIN0143 Group Coursework – Marking Scheme

Structure and Format (10%)

- Provides a concise summary.
- Coherent and appropriate structure.
- Adequate presentation, language, style, graphs, tables, and referencing.
- Appropriate use of visualisation

Methodology & Analysis (30%)

- Use of primary and/or secondary data.
- Rigorous analysis and interpretations.
- Alternative interpretations/arguments are considered.
- Appropriate and adequate methodology for topics.
- Detailed methodology facilitating replication of project and reproducibility of results.

Project Management [Trello] (15%)

- Progress made towards overcoming technical and operational challenges encountered during the project as a team.
- Good distribution of subtasks/task amongst team members.
- Good use of project management and communication tools.
- Clear which authors made what contribution to activity/analysis and overall submission

Quality of Narrative (30%)

- Demonstrates an understanding of business context.
- Topic is clearly identified and boundaries are asserted. Knowledge of relevant theories and their limitations.
- Limitations are identified and justified by reasonable arguments.
- Conclusions are fully consistent with evidence presented.
- Presents business recommendations.

Code / Markup / Referencing / Citations (15%)

- Code is well structured and clearly annotated.
- Markup is used correctly to aid understanding.
- Correct referencing of public code.
- Correct referencing of third party tools including Generative AI tools
- Demonstration of relevant literature and technical resources where appropriate.